

Lecture 6 Worksheet

Math 124, Fall 2026 at the University of Michigan

Some lectures will have an accompanying worksheet. We will do our best to include these problems directly in the notes as activities or worked examples.

Problem 1: Projections in $\mathbb{R}^2$

Let ℓ be the line $5x - 2y = 0$ and let $\vec{v} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$. We will find $\text{proj}_\ell(\vec{v})$, the orthogonal projection of $\vec{v}$ onto ℓ .

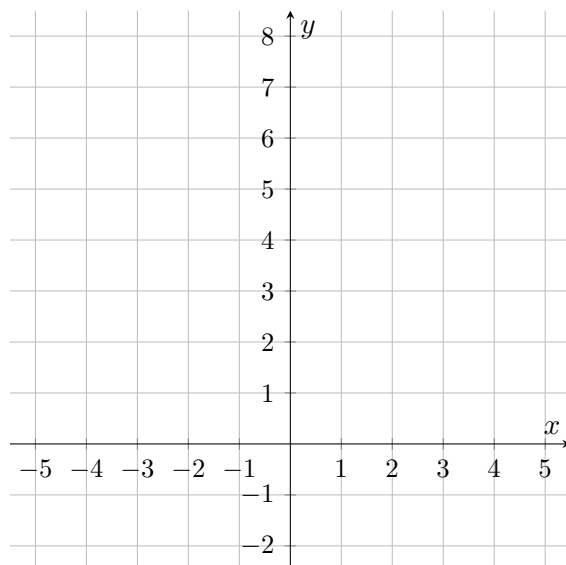
1. Draw $\vec{v}$ and ℓ below. Draw the orthogonal projection of $\vec{v}$ onto ℓ and roughly estimate $\text{proj}_\ell(\vec{v})$.
2. Pick any convenient nonzero vector $\vec{w}$ along ℓ and use

$$\text{proj}_\ell(\vec{v}) = \frac{\vec{v} \cdot \vec{w}}{\vec{w} \cdot \vec{w}} \vec{w}.$$

3. Pick a unit vector $\vec{u}$ along ℓ and use

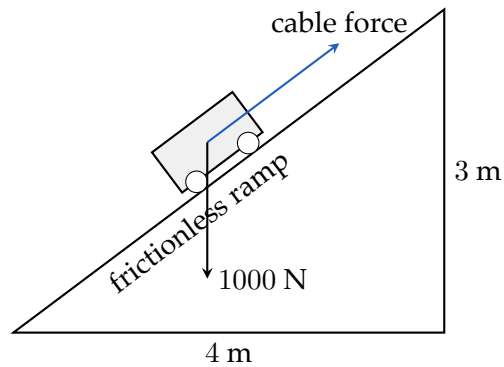
$$\text{proj}_\ell(\vec{v}) = (\vec{v} \cdot \vec{u}) \vec{u}.$$

You should get the same answer in parts 2 and 3. How does it match your estimate from part 1?



Problem 2: Vehicle on a Ramp

A vehicle with a weight of 1000 N rests on a frictionless ramp. For every 4 meters of horizontal distance, the ramp rises 3 meters. A cable parallel to the ramp is attached to the vehicle. How much force must the cable exert to keep the vehicle stationary (or move it at constant speed)?



Write the gravitational force as a vector, then break it into components parallel and perpendicular to the ramp. The cable must counter the parallel component.

1. Find a vector pointing upwards along the ramp: $\vec{w} =$

2. Write the gravitational force vector: $\vec{F}_g =$

3. Find its projection onto the line ℓ spanned by $\vec{w}$:

$$\text{proj}_{\ell}(\vec{F}_g) = \frac{\vec{F}_g \cdot \vec{w}}{\vec{w} \cdot \vec{w}} \vec{w}.$$

4. Write the cable force vector needed to counter this component. What is its magnitude?

5. If we used a unit vector $\vec{u}$ along ℓ instead of $\vec{w}$, how would we get the same answer?

Review: Geometry in $\mathbb{R}^2$

Before studying geometry in $\mathbb{R}^3$, recall the objects and ideas we have discussed in $\mathbb{R}^2$.

Geometric objects: lines through the origin, affine lines, vectors, and orthonormal bases.

Question (in English)	Answer (in Math!)
1. Describe a line ℓ through the origin using a nonzero vector $\vec{v}$ on ℓ .	$\ell = \text{span}(\vec{v}) = \{t\vec{v} : t \in \mathbb{R}\}$ (parametric form)
2. Describe an affine line ℓ' using a vector $\vec{v}_0$ on ℓ' and a vector $\vec{v}$ on ℓ (the parallel line through the origin).	$\ell' = \vec{v}_0 + \text{span}(\vec{v})$ $= \{\vec{v}_0 + t\vec{v} : t \in \mathbb{R}\}$ (parametric form)
3. What is the line through the origin perpendicular or orthogonal to ℓ called?	$\ell^\perp$
4. Write an equation for ℓ using a nonzero vector $\vec{w}$ on $\ell^\perp$.	$\vec{w} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0$
5. Write an equation for the affine line ℓ' using $\vec{v}_0$ and $\vec{w}$.	$\vec{w} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \vec{w} \cdot \vec{v}_0$
6. What is an orthonormal basis of $\mathbb{R}^2$?	A pair $(\vec{u}_1, \vec{u}_2)$ such that $\ \vec{u}_1\ = \ \vec{u}_2\ = 1$ and $\vec{u}_1 \cdot \vec{u}_2 = 0$.
7. Write an arbitrary vector $\vec{v}$ as a linear combination of an orthonormal basis $(\vec{u}_1, \vec{u}_2)$.	$\vec{v} = (\vec{v} \cdot \vec{u}_1)\vec{u}_1 + (\vec{v} \cdot \vec{u}_2)\vec{u}_2$
8. Project $\vec{v}$ orthogonally onto a line ℓ through the origin: find the vector on ℓ closest to $\vec{v}$.	$\text{proj}_\ell(\vec{v}) = (\vec{v} \cdot \vec{u})\vec{u} = \frac{\vec{v} \cdot \vec{w}}{\vec{w} \cdot \vec{w}}\vec{w}$, where $\vec{u}$ is a unit vector on ℓ and $\vec{w}$ is any nonzero vector on ℓ .

Now we'd like to move on to studying $\mathbb{R}^3$ (on the next page).

Geometry in $\mathbb{R}^3$

What objects should we consider, and what questions should we ask about them? Linear algebra studies objects that look “linear” or flat, such as lines and planes, and vectors.

Work in groups of 4–5 students. Feel free to use the whiteboards and blackboards to brainstorm. There is no single correct answer: we are assembling a list of questions!

Make a list of objects in $\mathbb{R}^3$ that one would like to study. Use the objects in $\mathbb{R}^2$ for inspiration.

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Are there more types of objects in $\mathbb{R}^3$ than in $\mathbb{R}^2$? Why?

Make a list of questions one might ask about these objects. List as many as you can; you do not need to answer them yet. Look back to $\mathbb{R}^2$ for inspiration.

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